

Celestial Boundary Toy Probes and Claim-Gated Physics Bridge: Phase 63.1 Release for HAOS-IIP

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Abstract

This paper freezes the Phase 57–63 celestial-facing sidecar arc in HAOS-IIP as a claim-gated toy-bridge release. The previous numbered paper stopped at Phase 56.5, where FMO telemetry was closed as a productive topology falsifier rather than a forced biological PASS. The present release resumes with a celestial-holography boundary audit, then adds a sequence of bounded toy/proxy probes: spherical harmonic boundary organization, synthetic soft-pole structure, toy displacement memory, ordered multi-pole supertranslation-like memory, and a composed supertranslation-to-memory response benchmark. The Phase 57 audit remains **OPEN**: HAOS-IIP has not recovered celestial holography, BMS charges, Virasoro/CCFT structure, an S-matrix dictionary, real soft theorems, or real gravitational memory observables. The toy/proxy rows are positive only inside their constructed benchmarks. The hardened spherical probe reports target score 0.983652 with margin 0.202698 over the best control. The toy soft proxy reports score 0.973247 with margin 0.250341. The toy memory probe reports score 0.970676 with margin 0.779772. The ordered multi-pole toy reports score 0.980025 with margin 0.228984. The composed shift-to-memory toy reports score 0.982918 with margin 0.455785. All five bounded toy/proxy rows have zero false-positive controls under their stated gates. This release is therefore a positive instrumentation result and a negative established-physics claim result at the same time.

1 Scope

This paper is release 63.1 in the HAOS-IIP numbered paper sequence. It packages the work performed after the 56.5 FMO topology-contrast closure note.

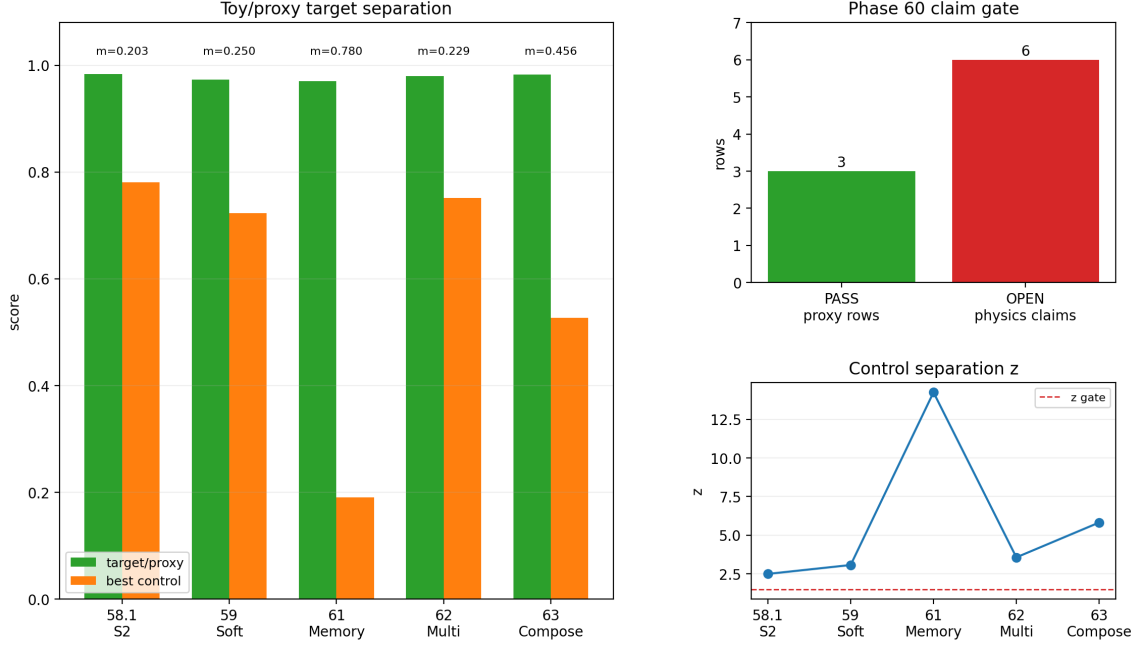
The implementation lives under:

`experiments/physics_bridge/`

The relevant sidecars are:

- `celestial_boundary_audit/`
- `spherical_harmonic_control_probe/`
- `soft_structure_proxy_test/`
- `claim_gated_bridge_update/`
- `gravitational_memory_toy_probe/`

HAOS-IIP 63.1: claim-gated celestial toy bridge summary



Bounded PASS rows are instrumentation checks only; celestial holography, BMS charges, Virasoro/CCFT, S-matrix reconstruction, real soft theorems, and real gravitational memory remain OPEN.

Figure 1: Phase 63.1 visual summary. Five bounded toy/proxy probes separate their targets from best controls, while the Phase 60 claim gate keeps established physics claims open. The PASS rows are instrumentation checks only.

- `multipole_supertranslation_probe/`
- `supertranslation_memory_composition_probe/`

No frozen HAOS-IIP core operator is modified by this release. The work is an external physics-bridge sidecar. Its purpose is to sharpen what can and cannot be said after several toy PASS results.

2 Where the Sequence Stopped

The previous numbered release was:

56.5 FMO Topology Contrast and Lessons for HAOS-IIP Biology Telemetry

That paper closed the first FMO telemetry arc as a diagnostic module. The important lesson was topology sensitivity: the same spectral telemetry instrument gave a robust microtubule toy PASS, but FMO remained a productive falsifier because pathway identity and control discrimination were not strong enough on a compact, delocalized transfer graph.

The present release moves to a different question:

Can HAOS-style spectral telemetry recognize controlled boundary-geometry and soft-like toy structures without promoting those toy structures into celestial holography claims?

3 Phase 57 Boundary Audit

Phase 57 deliberately turned the celestial-holography comparison into a boundary audit. Celestial holography starts from established asymptotic structures: BMS symmetry, soft theorems, gravitational memory, null infinity, S-matrix data, and celestial CFT correlators. HAOS-IIP does not contain those structures by default.

The audit result is:

Phase	Status	Failure mode
57	OPEN	CELESTIAL_REQUIREMENTS_NOT_CLOSED

This is not a weak result. It is the necessary boundary condition for the later toy probes. The later probes can pass only as controlled instruments, not as evidence that the full celestial dictionary has been recovered.

4 Toy and Proxy Probe Results

Phase	Probe	Target	Best control	Margin	False positives
58.1	Spherical harmonic boundary	0.983652	0.780955	0.202698	0.000000
59	Toy soft structure	0.973247	0.722906	0.250341	0.000000
61	Toy displacement memory	0.970676	0.190904	0.779772	0.000000
62	Multi-pole supertranslation toy	0.980025	0.751041	0.228984	0.000000
63	Shift-to-memory composition toy	0.982918	0.527133	0.455785	0.000000

The control-separation z values are:

Probe	Separation z
Spherical harmonic boundary	2.486753
Toy soft structure	3.064268
Toy displacement memory	14.250358
Multi-pole supertranslation toy	3.559799
Shift-to-memory composition toy	5.817631

5 Phase 58.1: Spherical Harmonic Boundary

The spherical-harmonic probe asks a narrow boundary-geometry question: can HAOS-style spectral telemetry distinguish a known sampled S^2 low-mode organization from generic smooth graph spectra?

The hardened 58.1 runner adds:

- low- ℓ subspace overlap;
- inter-band leakage checks;
- degeneracy and band-spacing checks;
- geodesic edge-weight signatures;
- edgewise distance-kernel fit checks;

- controls including coordinate permutation, weight shuffle, geodesic-bin weight shuffle, degree-preserving rewire, and ring-smooth graphs.

The result is **PASS**. The best control is `weight_shuffle_geodesic_control`. The interpretation is a known-target spherical-boundary sanity check only. It is not a recovery of null infinity or celestial boundary physics.

6 Phase 59: Toy Soft Structure

The soft-structure proxy constructs synthetic amplitude-like rows with known pole, residue, and factorization structure. The question is whether soft-specific telemetry detects residue, pole-location, and factorization breakage better than generic smoothness.

The result is **PASS**. The target soft-specificity score is 0.973247. The best control is the pole-location drift control, which scores 0.722906.

This is not a real soft theorem. It is a synthetic pole/residue/factorization benchmark.

7 Phase 60: Claim Gate

Phase 60 is the most important discipline layer in this arc. It makes the language boundary explicit:

Claim class	Status	Allowed reading
Spherical harmonic boundary geometry	PASS	Known-target S^2 mode sanity check
Toy soft-structure proxy	PASS	Synthetic pole/residue/factorization sanity check
Claim-boundary enforcement	PASS	Documentation hygiene
Celestial holography recovery	OPEN	Not addressed
BMS charge recovery	OPEN	Not tested
Virasoro / CCFT recovery	OPEN	Not tested
S-matrix reconstruction	OPEN	Not tested
Real soft theorem recovery	OPEN	Not tested
Gravitational memory observable	OPEN	Not tested by Phase 60

Phases 61–63 add more toy **PASS** rows, but they do not close any established physics claim. They extend the toy instrumentation ladder under the same gate.

8 Phase 61: Toy Displacement Memory

Phase 61 creates a synthetic permanent displacement-memory pattern on a sampled S^2 boundary. Controls include time shuffling, transient-only bursts, coordinate permutation, wrong low-mode signatures, smooth low-mode decoys, and stochastic drift.

The result is **PASS**:

Metric	Value
target memory specificity	0.970676
recoverability	0.999974
permanence	0.990143
temporal step	0.999998
mode signature	0.999998
best control	0.190904

This is a toy displacement-memory proxy. The real gravitational-memory observable row remains OPEN.

9 Phase 62: Multi-Pole Supertranslation Toy

Phase 62 extends the single memory target to an ordered $\ell = 2, 3, 4$ sequence of permanent mode shifts. The first run exposed a useful leak: the `event_order_shuffle_control` preserved the same final field and nearly the same multi-pole address, so final-field recovery alone was too permissive. The metric was hardened around ordered address-binding.

The final result is PASS:

Metric	Value
target supertranslation toy score	0.980025
final recoverability	0.999895
multipole address	0.999904
event order	0.987852
permanence	0.995698
best control	0.751041

The best control remains `event_order_shuffle_control`, as it should. It preserves much of the final field but fails the event-binding gate.

10 Phase 63: Supertranslation–Memory Composition Toy

Phase 63 composes the previous two ideas. A synthetic supertranslation-like shift occurs first. A later memory-like response is induced by that shift through a fixed toy response map in harmonic coefficient space.

This is stricter than testing the two fields independently. A control can preserve:

- the shift only;
- the memory only;
- the final field;
- smoothness;
- timing;
- or a wrong response field,

and still fail unless the recovered memory is predicted by the recovered shift under the fixed response map.

The result is PASS:

Metric	Value
target composition score	0.982918
shift recoverability	0.999969
memory recoverability	0.999800
composition relation	0.999853
causal order	0.994267
address binding	1.000000
best control	0.527133

The best control is `wrong_shift_right_memory_control`. It keeps the memory target but loses the shift-to-memory relation, which is exactly the intended failure.

11 Interpretation

The correct reading is:

HAOS-IIP now has a claim-gated celestial-facing toy bridge that can detect several controlled boundary and soft-like structures better than smoothness-preserving or partial-statistic controls.

The equally important negative reading is:

HAOS-IIP has not recovered celestial holography, BMS charges, Virasoro/CCFT structure, a real S-matrix, real soft theorems, or real gravitational memory.

Both statements must remain together. Separating them would break the point of this release.

12 Open Limits

- The spherical target is constructed and known.
- The soft-structure proxy uses synthetic amplitude-like tables, not physical amplitudes.
- The memory and supertranslation probes use toy fields on sampled S^2 , not detector memory observables.
- No BMS charge algebra is implemented.
- No Ward identity is tested.
- No Virasoro or celestial CFT operator algebra is recovered.
- No S-matrix data are reconstructed.
- No real gravitational-wave or scattering data are used.
- The controls are strong for the toy tasks, but they are not substitutes for physical amplitudes, asymptotic charges, or observational memory data.

13 Next Work

The clean next steps are:

- update the claim-gate table with Phase 61–63 toy rows as bounded proxy rows;
- run multi-resolution/asymptotic scaling across the spherical and memory toy families;
- add a first explicit toy Ward-identity benchmark before using any BMS language more strongly;
- keep all celestial-holography language under the Phase 57/60 gate until real asymptotic-symmetry or amplitude data are tested;
- return periodically to biology and dynamics so the project does not overfit to toy celestial benchmarks.

14 Release Artifacts

Primary artifacts:

- Phase 57 celestial boundary audit runner and generated requirement table.
- Phase 58.1 spherical harmonic control-probe runner, diagnostics, and figures.
- Phase 59 soft-structure proxy runner, diagnostics, and figures.
- Phase 60 claim-gated bridge table, report, and status JSON.
- Phase 61 gravitational-memory toy runner, diagnostics, and figures.
- Phase 62 multi-pole supertranslation toy runner, diagnostics, and figures.
- Phase 63 supertranslation–memory composition runner, diagnostics, and figures.
- Summary figure: `papers/figures/phase63_1_celestial_bridge_summary.png`.

The release PDF is:

63.1 Celestial Boundary Toy Probes and Claim-Gated Physics Bridge for HAOS-IIP.pdf